

Interacting with Masculinities: A Scoping Review

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ABSTRACT

Gender is a hot topic in the field of human-computer interaction (HCI). Work has run the gamut, from assessing how we embed gender in our computational creations to correcting systemic sexism, online and off. While gender is often framed around women and femininities, we must recognize the genderful nature of humanity, acknowledge the evasiveness of men and masculinities, and avoid burdening women and genderful folk as the central actors and targets of change. Indeed, critical voices have called for a shift in focus to masculinities, not only in terms of privilege, power, and patriarchal harms, but also participation, plurality, and transformation. To this end, I present a 30-year history of masculinities in HCI work through a scoping review of 126 papers published to the ACM Human Factors in Computing Systems (CHI) conference proceedings. I offer a primer and agenda grounded in the CHI and extant literatures to direct future work.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI theories, concepts and models; • **Social and professional topics** → User characteristics; Gender.

KEYWORDS

Masculinities, masculinity, gender, CHI, human-computer interaction, scoping review

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1 INTRODUCTION

Gender is on the frontlines of work aiming to raise attention to matters of inclusion, diversity, and social justice in human-computer interaction (HCI) [23, 135, 145, 178, 180, 191]. Gender is a multifaceted aspect of human identity and social organization [60, 69, 95, 96]. In line with emerging academic and cross-cultural consensus [95, 96], I approach gender as a social construct that is constituted, negotiated, and performed in a multitude of ways for a variety of functions within societies¹. Gender can be a mode of

expression, an internal self-identity, an external social category, and/or an abstract perception, even of objects [29, 69, 120, 222]. Gender is often framed as femininity and masculinity, but a wealth of work across time and cultures has challenged this “binary” model [60, 95, 96, 145, 178, 188, 198], including in HCI [178, 188, 191]. Indeed, HCI has a history of developing, studying, and critiquing technologies for gender inclusion and anti-sexism [12, 15, 165]. Feminist and intersectional HCI [178] projects have appeared alongside social movements like #MeToo and organizational changes, notably the Critical Computing, Sustainability, and Social Justice subcommittee² at CHI [38].

Technologies, especially computer-based ones, are implicated as sites and mediums of gender. Work has explored stereotypes in design decisions [43, 149, 155, 219], harassment in digital spaces [59, 125, 168], expert biases in recruitment and methods [114, 121, 145, 180, 187], and more. This work centres diversity and representation; highlights toxic behaviours towards women and genderful folk; and foregrounds the malignment of femininities and genderfulness. It has also unveiled the role of patriarchal systems [215] that centre, value, and privilege masculinities, where men and others uphold and benefit from the power offered to men by these systems. Much work has focused on limits and harms—but technology can also be a means of raising awareness and exploring gender expressions and experiences, as well.

Critical work has raised another challenge: framing [54, 107, 159, 209, 216]. When it comes to harms meted out through technology, women and genderful folk are rightfully centred. However, this can imply that sexism is under their purview alone. As Himmelsbach et al. [89:11] warn, “if women are studied solely, this might convey the impression [that gender] does not matter to men.” Men and masculinities are concealed [107], with men escaping responsibility or legitimately believing that they have no role to play [77, 209]. Others argue that women and genderful folk must change. A widely criticized [63, 92] instantiation of this is former Meta COO Sheryl Sandberg’s “lean in” feminism [174]. Some view the systems as too difficult to change or find that men do not participate [35, 174], perhaps viewing sexism as a personal problem [54, 209] or not knowing how to take action even when they want to [54, 159]. A more subtle frame is what *other* masculinities can be embraced, if not toxic ones—and how technology can help.

Work on masculinities in and outside of HCI may help reframe the situation and chart a path forward. A plurality of masculinities have been mapped out [107]. Masculinities intersect with other gender and sex identities [1, 40], as well as other factors, such as race [26, 139, 189] and sexuality [106]. As a social construct [69],

¹Gender is often contrasted with sex, which refers to the biological properties of people’s bodies, such as chromosomes, hormones, sex organs, and secondary sex

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characteristics, e.g., facial hair, that are categorized as male, female, and/or a range of intersexes [60, 95, 96]. As for gender, these categories are also social constructs. Also, the properties attributed to certain categories can vary widely within and across those categories; for example, breast size.

²Notably, the name of the subcommittee was updated in 2022 to explicitly include the “social justice” part.

masculinities are in flux, co-created, contested, and concretized—wittingly or otherwise—across cultures and over time [107]. Importantly, masculinities are not set in stone, nor are they the purview of men; we must all engage in interrogative, reflective, and practical work on masculinities, as a feature of research, at least. Indeed, deconstructing and diversifying the very notion of masculinities itself will be instrumental for progress on gender equality for everyone [107, 159]. We may approach masculinities as a facet of the user experience (UX), a demographic variable with a legacy of privilege within technology spaces [116, 212], and/or as a designable object [19, 224], one that may be shaped by computer technologies, such as social media [64, 73, 184]. Yet, how masculinities have been approached as a subject of study in HCI remains obscure. Moreover, most of us are not well-versed in theories of gender, as a matter of course in most forms of STEM education. This leaves us with a gap in our understanding of what has been done, what can be done, and what next steps should be taken, especially in HCI.

In this preliminary work, I sought to better understand whether and how masculinities have been approached within HCI. I asked a broad and exploratory question: *How have masculinities been approached in the field of HCI, if at all?* To this end, I carried out a scoping review [141, 156, 204] of the ACM Human Factors in Computing Systems (CHI) conference proceedings. I chose CHI as a comprehensive general venue featuring the highest quality of work in HCI. This work sets the stage for future primary research and systematic review work on masculinities within the field of HCI. As a retrospective, it can act as a stimulus for community self-reflection. It also addresses the need for more work centred on masculinities, in the plural, and how men can act in service of gender equality within HCI and the greater world.

2 METHODS

I conducted a scoping review [141, 156], a form of exploratory yet systematic literature review that aims to broadly capture a research subject, topic, or field of study [141, 156]. Scoping reviews are typically carried out before systematic reviews so as to identify the extent of the available primary research so far [141, 204], make a judgment on the value of carrying out systematic work [156], which has certain requirements and is much more resource-intensive [87, 151], and/or summarize the findings, trajectories, and gaps, especially when the work is novel or complex [204]. The value lies in tracing out histories, clarifying concepts, identifying knowns and unknowns (or even the unasked), and mapping out next steps [141, 156]. I used the PRISMA-ScR approach³ [204], a world-class standard [151] that helps maintain rigour when carrying out review work and provides a formal structure for reporting, ideal for ease of reading and peer review, as well as later meta-review work. While I undertook this project alone, I aimed to avoid bias in my procedure by employing the PRISMA-ScR. I registered this protocol before data collection on December 31st, 2022 at OSF⁴.

³Note that I deviated from the PRISMA-ScR to accommodate CHI reporting structures and HCI approaches to reporting, e.g., no structured abstracts.

⁴<https://osf.io/3kv7s>

2.1 Eligibility Criteria

All items, i.e., papers published to the CHI proceedings, that included masculinity as part of the work were accepted. If masculinity was referenced but not integrated into the work, e.g., in related work or future work, the item was excluded.

2.2 Information Sources and Search

The ACM Digital Library (ACM DL), the venue for the CHI proceedings, was searched on December 31st, 2022. The search query was: *AllFields:(masculinity) OR AllField:(masculinities)*. The results were filtered to the CHI proceedings.

2.3 Data Items

Metadata were extracted alongside: topic of study; research questions (RQs); HCI context, e.g., virtual reality (VR), hackerspaces; whether masculinities were central; whether gender was central; definition(s) of masculinities; whether these were explicit and quotable, implicit, such through associations of descriptors and masculinities, or unstated; whether masculinities were approached as a plural construct; whether gender was approached as a social construct; whether a binary approach to gender was taken; types of masculinities; theories; and citation(s) for all of these.

2.4 Selection Process and Data Synthesis

I downloaded the results of the query search from the ACM DL into Zotero and removed two invalid items (introductions to conference proceedings). I then randomly ordered and screened the items based on the eligibility criteria, removing five non-CHI papers. Next, I conducted a full-text review. This was done in three stages in parallel with data analysis, for which I used a reflexive thematic analysis approach [22]. This method is suitable for solo work; as Braun and Clark acknowledge, exploratory data analysis is subjective and relies on rater expertise; in my case, I am an experienced mixed methodologist in HCI who has published on gender. My process: First, I reviewed 20% of the items, and extracted data if eligible. At this stage, I developed the first set of codes based on patterns and highlights in the data. I then revisited the first 20% of items to refine the codes. I then reviewed and coded the next 20%. At this stage, I developed higher-level themes based on the expected contributions for scoping reviews but contextualized for the topic and field of HCI [141]: *social*, *design*, *research*, and *critique*. I then recoded all items. I removed 13 items that only referenced work or pointed to future work. I also categorized types of masculinities and theories. I used exploratory statistics where possible.

3 RESULTS

From an initial 146 records, 126 items between 1993–2022 were included (Figure 1). The data is available on OSF⁵. I now summarize the results. Counts and percentages were calculated against the total number of papers, unless specified.

⁵<https://osf.io/qgk8m>

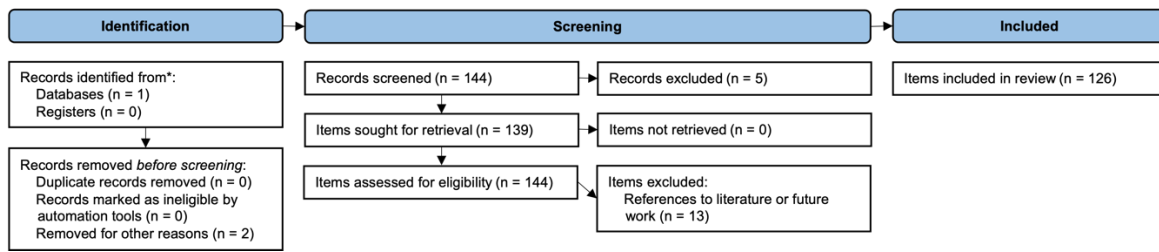


Figure 1: PRISMA flow diagram⁶ [151, 204] of the procedure for identifying items to be included and excluded.

3.1 Masculinity-Centrism in Topics and Contexts of Study

A variety of topics and 25 contexts were found. The most common contexts were general UI research, e.g., web and physical computing (17, 13%); HCI as a field, e.g., critiques of HCI practice, experiences of working in HCI, and bias in design, research, and practice (16, 13%); social media, e.g., Twitter and Reddit (14, 11%); and communities, e.g., online peer support, hackerspaces, and specific groups (13, 10%). Others included games, field work, crowdsourcing, VR, CUIs and voice UX, HRI and HAI, and design practice. Topics were so varied that it was difficult to summarize; I recommend reviewing the list in the OSF data set. I will draw out details and key into this diversity in the following sections.

Despite the target of this work and search terms used, only 14 (11%) papers had masculinities as a main topic. Additionally, about half or 65 (52%) papers were focused on gender. Examples of topics include: gender norms and attitudes in father blogs [124]; African American men and computing identity [98]; heteronormativity and hegemonic masculinity in VR porn [227]; sexist beliefs via sexualization/objectification of avatars within gaming contexts [21]; and urinal games [133]. Topics cover a variety of masculinities, intersectionality, identity and body, and sexism.

Most other papers had masculinities as subfactors or emergent factors. Martens [131] used a scale to evaluate a novel statistics interface that included the subfactors “feminine,” “masculine,” and “unisex.” Williams and Gilbert [223:3] contextualized their case studies against the scholarship of feminist and critical disability studies to point out the role that certain masculinities within certain cultural contexts play in power imbalances within peer review processes: “This self-invisibility is the specifically modern, European, masculine, scientific form of the virtue of modesty . . . modern fantasies of objectivity have historically rendered the white male as the invisible default witness to scientific fact.”

Masculinities were also approached at a meta level, in relation to their design and/or research praxis. Schechter, Egelman, and Reeder [176] explained that they used masculine and feminine pronouns in the accounts of their social-authentication system “for clarity,” a choice that assumes a reader would be confused by the use of the same pronouns for different characters, as well as one that belies gender binary thinking. Tachtler et al. [196] provided a nuanced discussion about recruitment, noting that they had aimed but were unable to recruit a diverse sample, and considering valid reasons why, i.e., most unaccompanied migrant youth in and around their site were young men. Kao et al. [103] argued that their selection

of stereotypical “male” and “female” voiced avatars was meant to achieve ecological validity against the most common setups in real games, noting that “a binary view of gender is problematic” [103:7].

3.2 Definitions of Masculinities

Only four papers (3%) explicitly operationalized masculinity or masculinities. These were sourced from critical studies, dictionaries, cultural studies, or were unsourced. I outline them below:

- Rubin, Blackwell, and Conley in 2020 [168:2], citing Coston and Kimmel [41]: “the behaviors and expectations culturally associated with boys and men.”
- Pater et al. in 2019 [153:2-3] combine “masculinity” and “male,” and “male” and “man,” operationalizing these as identity expression through social media, notably by way of “external appearance.” They use the Oxford English Dictionary to define “male,” but refer to the dictionary term of “man”: “having qualities or appearance traditionally associated with men, especially strength and aggressiveness ... gender as “spectrum” [233].
- Danieleescu and Christian in 2018 [44:6], citing Hofstede’s “cultural dimensions,” or construction of gender as a culture-wide, general attribute [90]: “Masculinity, contrasted with femininity, is sometimes expressed as ‘tough vs. tender’ - it quantifies how competitive a society is, and social rewards for achievement vs. cooperation.”
- Dosono and Semaan in 2018 [51:5] created a thematic code for “masculinity” in the context of American Asians and Pacific Islanders (AAPIs) on Reddit, defining it as: “Critiques of qualities traditionally associated with men.”

In 14 cases (11%), definitional qualities were implied by association. A word frequency analysis of 468 terms from the sentences in which “masculinity/ies” was found highlights several commonalities: *gender* (11), *feminine*, (7), *associated*, *home*, *male*, *men*, *physical*, *qualities*, *traditionally*, *work* (4 each), *binary*, *body*, *competitive*, *data*, *design*, *female*, *gaming*, *individuals*, *psych*, *strength*, *traits*, *violence* (3 each), *activities*, *agentic*, *aggressive*, *alternative*, *assertive*, *categories*, *characterize*, *clothing*, *consistent*, *different*, *domestic*, *dominant*, *emotions*, *express*, *form*, *guru*, *identifying*, *identity*, *implicit*, *labor*, *man*, *men’s*, *minorities*, *models*, *people*, *power*, *pressure*, *sir*, *sport*, *technical*, *technology*, *traditional*, *trans*, *transmasculine*, *transmen*, *type*, *user*, *women working* (2 each). This indicates that masculinities were positioned against femininities and women. Qualities align with common views, such as strength, violence, dominance, tradition . . .

and technology. Still, a portion point to alternative models, intersectional factors, and trans identities.

81 papers (64%) did not define or operationalize masculinity directly or indirectly. A word frequency analysis of 825 terms from the sentences in which “masculinity/ies” was found reveals: *gender* (26), *feminine* (19), *men* (18), *women* (14), *male* (11), *fem*, *participants*, *technology* (8 each), *binary*, *female*, *two* (7 each), *choices*, *Hofstede* (6 each), *abuse*, *norms*, *people*, *sex*, *white* (5 each), *avatar*, *boys*, *communities*, *control*, *create*, *cultural*, *experiences*, *gendered*, *modern*, *perceived*, *play/games*, *traditional* (4 each), *culture*, *dimensions*, *dominant*, *environment*, *hegemonic*, *identity*, *majority*, *males*, *neutral*, *options*, *others*, *participant*, *pronouns*, *role*, *social*, *trans*, *work*, *young* (3 each). While this largely matches the above, the work of Hofstede and cultural-level operationalizations of masculinity/ies are highlighted. Moreover, there is mention of male-as-neutral and specific reference to games/play. Finally, the intersectional factors and trans identities are less present, which we might expect since these tend to be marginalized and thus at risk of being overlooked.

Nearly half or 60 papers (48%) took a pluralistic approach to masculinities. Still, 8 papers (6%) took a singular approach, and for more than half (58, 46%) it was unclear. For example, Madden et al. [126:1] cited literature on how gamers and gaming cultures have been characterized, without making it clear how they felt about it: “male and female gamers play competitive games in roughly equal numbers . . . esports . . . are still viewed as ‘male-dominated’ . . . the masculine and feminine cultures in gaming are still surrounded by commonplace assumptions, such as males being ‘aggressive’ and ‘undesirable’ individuals.” Merely citing literature does not necessarily imply the authors’ own stances. Most papers (92, 73%) approached masculinities as a constructed object, although it was hard to judge in 32 (25%) of cases. For example, Gonzales and Fritz [76] talked to folks engaging in crowdsourcing to fund top survey or “reconstruction of a masculine chest” [76:2371], recruiting “only transmen or those who identified as transmasculine” [76:2372]. This points to plurality as well as the intersections of identity, positionality, and personal choice. Still, nearly half (58, 46%) relied on a gender binary perspective, with 54 (43%) going beyond the binary and 14 (11%) unclear. Given the clear distribution here, I ran a Kendall tau-b test, finding a significant negative relationship between year of publication and binary positioning, $\tau_b = -.224$, $p = .006$. This indicates that the use of a binary framing has declined over the years, though it is still present.

In short, men and masculinities were largely taken for granted. When defined, “masculinity” was characterized in a range of ways: cultural norms and behaviours, identity and appearance, an attribute of cultures as a whole, a critique of traditional qualities associated with men and boys. While many have taken a binary framing, there appears to be a trend away from this and especially towards constructivist and pluralistic characterizations—even if the authors do not address this directly, such as with definitions or references to theoretical frameworks.

3.3 Types of Masculinities

I now review the types of masculinities invoked and defined in the literature. This is not an inclusive classification of masculinities,

nor does each category comprise an exhaustive list of types. This represents the current state of affairs.

3.3.1 General Types of Masculinities. Types of masculinities named across the corpus of papers, in order from most common to least:

- *Hypermasculinity*, described as “culture of college fraternities” ... “barbarian, manly hero” ... “dominance, violence, and lack of emotional expression” [21, 25, 58, 82, 99, 126], with references to Witkowski [225] and Zolides et al. [232].
- *Hegemonic masculinity*, or the dominant, singular model of masculinity in a given society, which was not referenced, but used in several papers [2, 99, 118, 190, 227].
- *Toxic masculinity*, also not defined or sourced but used in several papers [25, 112, 142, 146, 168].
- *Normative masculinity*, referring to adherence to a given society’s expectations for men and masculinities, described as “being assertive, demonstrating bravery through risk-taking, upholding heterosexuality and rejecting femininity, and establishing dominance through aggression” ... “appreciating and practising sports” [168, 211], with references to Mahalik et al. [127] and Pascoe and Bridges [152].
- *Alternative masculinities*, a pluralistic concept where “men are able to express their emotions, reject violence, and champion fighting all forms of oppression of women and other men” ... a “softer” form of masculinity” [168, 211], referencing Pascoe and Bridges [152].
- *Rugged masculinity*, described as “taming ‘virgin’ nature, the problems of habitation by indigenous peoples, and the issues of the supernatural associated with the encounter with wilderness” ... “‘rugged individualism’ culture in computing” [146, 185], referencing Dourish [53], Ensmenger [57], Fox and Tang [65], and Salter and Blodgett [172].

Others include fragile masculinity [168], masculinity anxieties [168], male-default values [125] traditional masculinity [66], *babak* (the Javanese version of hegemonic masculinity) [118], and supportive masculinity [173].

3.3.2 Technology-Oriented Masculinities. The two most common forms of technology-oriented masculinities were:

- *Geek masculinity*, a general term variably described as a “masculine understanding of identity that is visible across technology culture” ... “in which technological mastery forms the basis of masculine esteem and social status” [25, 67, 136], with references to Kendall [105], Bucholtz [27], Eglash [55], and Lin and den Besten [119].
- *Toxic gamer culture*, focused on video game sites and which “frames gaming as a male-gendered, potentially violent space” [82, 126], via Consalvo [39].

Others included brogrammers [25], alpha and beta masculinities [25], masculine technophile [190], technology czars and gurus [190], and masculine prototypicality in technology [162].

3.4 Theories and Frameworks of Masculinities

Most (28, 92%) theories and frameworks referenced by authors in this corpus of work were general or non-disciplinary: identity intersectionality [71, 91], construction [75, 122], and multiplicity [28]; masculine norms [84, 163], traits [17, 179, 186], roles

Table 1: Thematic framework of approaches to masculinities at CHI

Theme	Code	Papers	Count (%)
Social	Behaviour	[2, 3, 5, 6, 11, 14, 15, 20, 25, 36, 45, 47, 51, 52, 58, 62, 66, 68, 76, 80, 83, 94, 97, 103, 109, 114, 117, 123, 125, 126, 128, 133, 134, 136, 138, 140, 142, 146, 147, 149, 153, 154, 158, 162, 164, 167, 168, 170, 177, 180, 181, 183, 190, 192–195, 199, 200, 202, 208, 211, 217, 218, 221, 223, 227, 228, 230, 231]	69 (55%)
	Attitudes	[2–6, 11, 14, 16, 20, 21, 25, 30, 34, 36, 42–45, 47, 50–52, 58, 62, 66–68, 76, 78, 80, 83, 88, 97, 98, 103, 104, 108–110, 112, 123–126, 128, 130–132, 134, 136, 138, 140, 142, 143, 146–149, 153, 154, 158, 167, 170, 173, 175, 177, 183, 190, 192, 195–197, 199, 200, 202, 205, 211, 218, 221, 227, 228, 230]	83 (66%)
	Identity	[2–6, 11, 16, 21, 25, 31, 33, 36, 51, 52, 58, 66–68, 70, 76, 79, 83, 94, 98, 99, 103, 110, 117, 125, 126, 128, 134, 138, 140, 142, 146, 147, 153, 154, 158, 177, 188, 190, 192, 195, 197, 199, 200]	50 (40%)
Design	Agent Attribute	[11, 21, 44, 50, 68, 70, 72, 88, 102, 103, 109, 111, 117, 123, 129, 150, 180, 182, 194, 205, 221, 230]	22 (17%)
	Interface Pattern	[3, 14, 30, 61, 66, 72, 78, 104, 108, 110, 131, 132, 134, 137, 138, 143, 147, 150, 167, 171, 202, 211, 217, 229]	24 (19%)
	Experience	[2–4, 11, 14, 16, 20, 21, 25, 31, 33, 34, 36, 45, 47, 51, 52, 58, 62, 66, 76, 80, 83, 94, 97, 103, 109, 110, 113, 114, 123, 125, 126, 130, 132–134, 136, 138, 140, 142, 143, 146, 147, 158, 162, 167, 170, 173, 183, 190, 193, 197, 199, 200, 202, 208, 218, 227, 228, 230, 231]	62 (49%)
Research	Space	[2, 4–6, 11, 14, 16, 20, 25, 31, 34, 36, 42, 45, 45, 47, 51, 52, 66, 67, 72, 76, 80, 83, 94, 97, 98, 113, 124–126, 130, 133, 134, 136–138, 140, 142, 143, 146, 153, 154, 158, 162, 164, 167, 170, 173, 183, 185, 190, 192, 194, 196, 199, 200, 208, 218, 227, 230, 231]	62 (49%)
	Method	[12, 31, 43, 50, 66, 67, 88, 129, 148, 149, 164, 167, 180, 211]	14 (11%)
	Reporting	[99, 115, 176, 180, 188, 203, 208]	7 (6%)
Critique	Method	[4, 23, 30, 42, 72, 88, 89, 99, 128, 148, 154, 167, 175, 177, 180, 181, 188, 195, 223]	19 (15%)
	Reporting	[23, 89, 99, 109, 126, 154, 180, 188, 223]	9 (7%)
	Technology	[3, 14, 23, 30, 31, 33, 36, 68, 79, 81, 82, 103, 112, 118, 125, 134, 136, 138, 149, 162, 167, 170, 180, 192, 193, 196]	26 (21%)
	Field	[2, 10, 12, 23, 42, 47, 51, 83, 89, 99, 118, 129, 130, 136, 146, 164, 167, 170, 173, 177, 185, 188, 190, 192, 195, 200, 202, 223]	28 (22%)

[127], and cultures [166]; social construction [60, 69]; performativity [37, 213, 214]; gender schema theory [18]; gender rules [210] and ideology [46]; Hofstede’s masculinity index for cultures [90]; othering [144]; heteronormativity, heteropatriarchy [91], and value neutrality [9]; gender role strain paradigm [160, 161]; father involvement [85]; muscle dysmorphia [157] and bigorexia [74]; male-dominated [65, 93, 226] and masculinized spaces [100]; autonomy from masculinity [170]; androcentrism and male-as-default [48]; alpha male effect [86]; and masculine disclosure [172]. Others (6, 18%) were tech-oriented: gender-agnostic platforms [125]; cultural stereotypes as gatekeepers [32] and men’s/boy’s clubs [99, 169, 201]; online disinhibition effect [207]; the manosphere [73]; co-production of gender and tech; and data feminism [49] (data as masculine, i.e., rational and objective).

3.5 Summarizing Approaches to Masculinities at CHI: Social, Design, Research, Critique

I summarize the state of affairs across this 30-year corpus of CHI papers with the themes and codes in Table 1.

4 DISCUSSION

Global shifts in how “masculinity” is viewed, personally and politically, are taking place alongside worldwide calls for action on gender bias and sexism in the technosphere. How has CHI risen

to the challenge? While small, the pool of work focusing on or including a component of masculinities represents a diverse array of work. Still, there are gaps and potentials not yet traversed that may be especially suitably for HCI work, if not CHI specifically.

4.1 Bringing in Masculinities from the Extant Literature

We can use the extant literature in men and masculinities studies and gender studies to seed new directions. Drawing on my expertise, I provide this curated list of influential work as a starting point.

- *Technomascularity* [101] refers to how men portray themselves as advanced computer users and rely on this portrayal when relating to others. Here, computer use and especially mastery is taken on as a social identity. In HCI work, technomascularity may guide: the selection of participants based on gender and/or technology-oriented identities, especially in multi-user contexts; the design of questionnaires and other probes that involve self-reports of technical identity, ability, and/or experience; and observational and analysis frameworks at sites and in data where technical mastery may play a role. For example, gendered self-selection of peer programming teams in a classroom setting may be understood through a technomascularity framework. Technomascularity

relates to geek masculinity, found in this survey across several papers. Technomascularity is about technical mastery as a form of power, while geek masculinity is an alternative when mastery of dominant forms of masculinity are perceived to be unachieved or unachievable.

- *Inclusive masculinity theory (IMT)* [7] was “developed to explain sport and fraternity settings where the social dynamics were not predicated on homophobia, stoicism or a rejection of the feminine” [8:549]. As this review has shown, technology spaces, activities, and roles across HCI contexts carry a range of masculine-centric or -dominant characteristics, values, and demarcations. Even the digital instantiation of sports, esports, has been explored [126]. If forms of masculinity trickle down from the larger culture, we may expect to find similarities when comparing to other masculine-centric or -dominant domains, like sports and fraternities. IMT presents a more nuanced framing to further shape these expectations and push us to consider alternatives. For example, we may be primed to look for certain forms of gendered interlocations between a self-identifying jock and a stereotypically deferential feminine-voiced virtual assistant. We may miss or decenter engagements that do not fit expectations, such as the jock taking on emotional labour for a friend by searching about a seemingly non-gendered, benign topic, like the closest store to get a prepaid phone card.
- *Hybrid masculinities* are defined as “men’s selective incorporation of performances and identity elements associated with marginalized and subordinated masculinities and femininities [24:246]. HCI now offers a wealth of ways in which to express, play with, challenge, deconstruct, and reify gender through identity and performance beyond traditional text modalities. We can choose and customize avatars; we can modify our appearance in realtime on Zoom; we can change our voice with vocalization software; we can even produce deepfakes of ourselves and others. A key element of this is *change*: we are not stuck with a single mode of expression. Hybrid masculinities could illuminate and explain longitudinal engagements with technologies as modes of self-expression. Social media, video games, wearables at cosplay events . . . there are many HCI-oriented sites where hybrid masculinities could be found.
- *Caring masculinities* are defined by anti-domination, positive emotion, interdependence, and a focus on relations [56]. It may be employed as an alternative to the manosphere already explored at CHI [73]. Investigations could be as general as how people who identify as men or masculine conduct themselves in interpersonal exchanges online to specifically tracing out communities and movements centred on forms of caring masculinities, including and beyond anti-misogyny initiatives. We can also revisit the gaming and VR spaces found in the reviewed work to explore interactive narratives, characters, and mechanics that allow people to take on caring masculinist personas and modes of engagement.
- *Flexible, strategic* [13], and *chameleon* [220] masculinities refer to “code-switching” in masculinity performance. Do those who identify as men or masculine switch between modes of

expression, even suddenly or in rapid succession, when interacting with certain others or moving between technology spaces? Could such “code-switching” be embodied in virtual characters, agents, and robots designed with masculine cues? There is much to explore on either side of the HCI equation.

- *Postcolonial masculinities* are those that centre masculinities beyond Western and “First World” contexts [189]. In this review, only one was found: the Javanese *bapak* [118]. Initiatives in the field of HCI and especially at CHI have highlighted and pressed for recognition and engagement on matters of diversity, equity, and inclusion beyond gender. Could postcolonial masculinities complement the work this review has found on rugged masculinities [146, 185], for example? This also means taking an intersectional lens to gender and in this case masculinities. For instance, do “caring masculinities” look the same in a South Asian WhatsApp group chat compared to a British one? We should be cautious about making assumptions and leaning on generalizations from WEIRD research at CHI [121] and adjacent spaces, such as HRI [234]. Still, HCI is a radical, creative, and political space, welcoming of inclusive knowledge and change in praxis. Ideas run the gamut. For instance, we could create Two-Spirit avatars and modes of engagement in interactive stories and video games that link masculinities, femininities, and gender identities and expression to sexuality and socio-cultural roles and hierarchies that do not necessarily map on Western LGBTQI+ models and distinctions [235].

4.2 An Agenda for Future Interactions with Masculinities

I offer a non-exhaustive list of ideas and prompts based on the surveyed work and summarizing thematic framework.

- *Social: Hack Masculinities with New Forms of Education and Events.* Masculinities come to bear in social HCI contexts, with most work covered in this review falling within attitudinal (66%), behavioural (55%), experiential (49%), and spatial (49%) themes. A complementary, triangulated approach could be explored at these intersections in the forms of educational initiatives and social events. Plank [159] asks why we have tech events for girls but we do not have nursing events for boys. Could we design games that explore multiple forms of masculinities? What about VR applications that allow men and boys to freely explore gender expression? Could we design hackathons on technologies that confront toxic masculinity . . . encourage caring masculinities . . . or involve “feminine” activities, like digital sewing?
- *Design: Create New Prototypes from the Lenses of Extant Theories.* While the body of work covered in this review drew on several general (3.3.1) and HCI-oriented frameworks of masculinities (3.3.2), I point out several more candidates that have not yet been explored (4.1). How can we use these theories in HCI? The paths forward are too numerous to name, but I can offer a few more ideas. How could HCI approach, for example, postcolonial masculinities, especially given the decolonizing work [154, 206] already underway? Are there

caring masculinities in the e-health space or in mental health and wellbeing spaces on social media? Could there be?

- *Research: Explore a Diversity of Theories.* As the content analysis results (3.2) indicate, much of the work covered in this review appears to speak to theories, even if those theories were not used. Can we revisit previous work—even just the data for further analysis—in case instances of novel (for CHI) forms of masculinities were missed? Can we offer our data as part of a new open science initiative to uncover implicit connections to existing theoretical frameworks? Doing so would not only be useful for the field of HCI, but also feed back into larger knowledge and theoretical bases of gender and masculinities. This could show how CHI as a venue contributes to general knowledge in a concrete way. This would also reveal the how HCI is distinct, leading to offshoot theories and potentially new ideas for design and research.
- *Critique: Let's Be Reflexive and Change Our Ways.* We lean on masculinities, whether we are conscious of it or not. Still, only 3% of the corpus operationalized “masculinities,” with 64% leaving the concept undefined and nearly half (48%) taking an unclear stance, even for recruiting and/or demographics reporting. Still, the other half (48%) has taken on a pluralistic approach, even without providing a clear definition or using a theoretical framework. We may be at a key juncture for reflexivity [164] as a field of study and practice. CHI can lead the way. Change can be small, such as rethinking how we ask about gender for demographics [187]. How do we wish to interact with masculinities?

4.3 Limitations

This work was limited by the focus on the CHI conference; future work should scope out the literature in other HCI venues, including conferences, journals, and other venues. I alone carried out this work; while aimed for self-correction by remaining reflexive and employing the PRISMA-ScR, I acknowledge that the codings and classifications could be limited by this solo approach. Future work can test and expand these frameworks with multiple raters.

5 CONCLUSION

In this scoping review, I have traced out a history of masculinities at the premier international CHI conference. I have extended the base offered by this body of work by weaving in extant theories and highlighting trajectories for future work. HCI spaces like CHI have much to offer, in the past, present, and future, by approaching masculinities as a matter of inclusion, diversity, and social justice for everyone.

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REFERENCES

- [1] Miriam J. Abelson and Tristen Kade. 2019. Trans masculinities. In *Routledge International Handbook of Masculinity Studies*. Routledge, 165–173. <https://doi.org/10.4324/9781315165165-16>
- [2] Michael Ahmadi, Rebecca Eilert, Anne Weibert, Volker Wulf, and Nicola Marsden. 2020. Feminist Living Labs as Research Infrastructures for HCI: The Case of a Video Game Company. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–15. <https://doi.org/10.1145/3313831.3376716>
- [3] Alex A. Ahmed. 2019. Bridging Social Critique and Design: Building a Health Informatics Tool for Transgender Voice. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–4. <https://doi.org/10.1145/3290607.3299077>
- [4] Taghreed Alshehri, Norah Abokhodair, Reuben Kirkham, and Patrick Olivier. 2021. Qualitative Secondary Analysis as an Alternative Approach for Cross-Cultural Design: A Case Study with Saudi Transnationals. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445108>
- [5] Tawfiq Ammari, Sarita Schoenebeck, and Daniel M. Romero. 2018. Pseudonymous Parents: Comparing Parenting Roles and Identities on the Mommit and Daddit Subreddits. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3173574.3174063>
- [6] Nazanin Andalibi, Oliver L. Haimson, Munmun De Choudhury, and Andrea Forte. 2016. Understanding Social Media Disclosures of Sexual Abuse Through the Lenses of Support Seeking and Anonymity. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 3906–3918. <https://doi.org/10.1145/2858036.2858096>
- [7] Eric Anderson. 2009. *Inclusive Masculinity: The Changing Nature of Masculinities*. Routledge, New York. <https://doi.org/10.4324/9780203871485>
- [8] Eric Anderson and Mark McCormack. 2018. Inclusive Masculinity Theory: overview, reflection and refinement. *Journal of Gender Studies* 27, 5, 547–561. <https://doi.org/10.1080/09589236.2016.1245605>
- [9] Lindsey Andrews. 2015. Breathing Race into the Machine: The Surprising Career of the Spirometer from Plantation to Genetics by Lundy Braun (review). *Configurations* 23, 1, 127–130. <https://doi.org/10.1353/con.2015.0000>
- [10] Matthew P. Aylett and Shaun Lawson. 2016. The Smartphone: A Lacanian Stain, A Tech Killer, and an Embodiment of Radical Individualism. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 501–511. <https://doi.org/10.1145/2851581.2892581>
- [11] Jeffrey Bardzell, Shaowen Bardzell, Guo Zhang, and Tyler Pace. 2014. The Lonely Raccoon at the Ball: Designing for Intimacy, Sociability, and Selfhood. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 3943–3952. <https://doi.org/10.1145/2556288.2557127>
- [12] Shaowen Bardzell. 2010. Feminist HCI: Taking stock and outlining an agenda for design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*, 1301–1310. <https://doi.org/10.1145/1753326.1753521>
- [13] Adina Batnitzky, Linda McDowell, and Sarah Dyer. 2009. Flexible and Strategic Masculinities: The Working Lives and Gendered Identities of Male Migrants in London. *Journal of Ethnic and Migration Studies* 35, 8, 1275–1293. <https://doi.org/10.1080/13691830903123088>
- [14] Laura Beckwith, Cory Kissinger, Margaret Burnett, Susan Wiedenbeck, Joseph Lawrance, Alan Blackwell, and Curtis Cook. 2006. Tinkering and Gender in End-User Programmers' Debugging. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 231–240. <https://doi.org/10.1145/1124772.1124808>
- [15] Rosanna Bellini, Angelika Strohmayr, Ebtsam Alabdulqader, Alex A. Ahmed, Katta Spiel, Shaowen Bardzell, and Madeline Balaam. 2018. Feminist HCI: Taking stock, moving forward, and engaging community. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18)*, 1–4. <https://doi.org/10.1145/3170427.3185370>
- [16] Rosanna Bellini, Alexander Wilson, and Jan David Smeddinck. 2021. Fragments of the Past: Curating Peer Support with Perpetrators of Domestic Violence. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445611>
- [17] Sandra L. Bem. 1974. The measurement of psychological androgyny. *Journal of Consulting and Clinical Psychology* 42, 2, 155–162. <https://doi.org/10.1037/h0036215>
- [18] Sandra Lipsitz Bem. 1981. Gender schema theory: A cognitive account of sex typing. *Psychological Review* 88, 354–364. <https://doi.org/10.1037/0033-295X.88.4.354>
- [19] Anne-Jorunn Berg and Merete Lie. 1995. Feminism and constructivism: Do artifacts have gender? *Science, Technology, & Human Values* 20, 3, 332–351. <https://doi.org/10.1177/016224399502000304>
- [20] Mark Blythe, Andrew Monk, and Jisoo Park. 2002. Technology Biographies: Field Study Techniques for Home Use Product Development. In *CHI '02 Extended Abstracts on Human Factors in Computing Systems*, 658–659. <https://doi.org/10.1145/506443.506532>
- [21] Jason T. Bowey, Ansgar E. Depping, and Regan L. Mandryk. 2017. Don't Talk Dirty to Me: How Sexist Beliefs Affect Experience in Sexist Games. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 1530–1543. <https://doi.org/10.1145/3025453.3025563>
- [22] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2, 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [23] Samantha Breslin and Bimlesh Wadhwa. 2015. Towards a Gender HCI Curriculum. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts*

- on *Human Factors in Computing Systems*, 1091–1096. <https://doi.org/10.1145/2702613.2732923>
- [24] Tristan Bridges and C. J. Pascoe. 2014. Hybrid Masculinities: New Directions in the Sociology of Men and Masculinities. *Sociology Compass* 8, 3, 246–258. <https://doi.org/10.1111/soc4.12134>
- [25] Sian JM Brooke. 2022. Nice Guys, Virgins, and Incels: Gender in Remixing and Sharing Memes at Hackathons. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3517627>
- [26] Mary Bucholtz. 1999. You da man: Narrating the racial other in the production of white masculinity. *Journal of Sociolinguistics* 3, 4, 443–460. <https://doi.org/10.1111/1467-9481.00090>
- [27] Mary Bucholtz. 1999. “Why Be Normal?”: Language and Identity Practices in a Community of Nerd Girls. *Language in Society* 28, 2, 203–223.
- [28] Judith Butler. 1988. Performative acts and gender constitution: An essay in phenomenology and feminist theory. *Theatre Journal* 40, 4, 519–531.
- [29] Judith Butler. 2004. *Undoing Gender*. Routledge. <https://doi.org/10.4324/9780203499627>
- [30] Marta E. Cecchinato, Anna L. Cox, and Jon Bird. 2017. Always On(Line)? User Experience of Smartwatches and Their Role within Multi-Device Ecologies. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 3557–3568. <https://doi.org/10.1145/3025453.3025538>
- [31] Le Chen, Ruijun Ma, Anikó Hannák, and Christo Wilson. 2018. Investigating the Impact of Gender on Rank in Resume Search Engines. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3173574.3174225>
- [32] Sapna Cheryan, Allison Master, and Andrew N. Meltzoff. 2015. Cultural stereotypes as gatekeepers: increasing girls’ interest in computer science and engineering by diversifying stereotypes. *Frontiers in Psychology* 6. Retrieved January 16, 2023 from <https://www.frontiersin.org/articles/10.3389/fpsyg.2015.00049>
- [33] Toby Chong, Nolwenn Maudet, Katsuki Harima, and Takeo Igarashi. 2021. Exploring a Makeup Support System for Transgender Passing Based on Automatic Gender Recognition. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445364>
- [34] Shaan Chopra, Rachael Zehrung, Tamil Arasu Shanmugam, and Eun Kyoung Choe. 2021. Living with Uncertainty and Stigma: Self-Experimentation and Support-Seeking around Polycystic Ovary Syndrome. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445706>
- [35] Donna Chrobot-Mason, Jenny M. Hoobler, and Jasmine Burno. 2019. Lean In Versus the Literature: An Evidence-Based Examination. *Academy of Management Perspectives* 33, 1, 110–130. <https://doi.org/10.5465/amp.2016.0156>
- [36] Tya Chuanromanee and Ronald Metoyer. 2021. Transgender People’s Technology Needs to Support Health and Transition. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445276>
- [37] Cynthia Cockburn and Ruža First-Dilić. 1994. *Bringing Technology Home: Gender and Technology in a Changing Europe*. Open University Press Buckingham.
- [38] Rob Comber, Shaowen Bardzell, Jeffrey Bardzell, Mike Hazas, and Michael Muller. 2020. Announcing a new CHI subcommittee: Critical and sustainable computing. *interactions* 27, 4, 101–103.
- [39] Mia Consalvo. 2012. Confronting toxic gamer culture: A challenge for feminist game studies scholars. <https://doi.org/10.7264/N33X84KH>
- [40] Anne Constantinople. 2005. “Masculinity-femininity: An exception to a famous dictum?” *Feminism & Psychology* 15, 4, 385–407. <https://doi.org/10.1177/0959-353505057611>
- [41] Bethany M. Coston and Michael Kimmel. 2012. Seeing Privilege Where It Isn’t: Marginalized Masculinities and the Intersectionality of Privilege. *Journal of Social Issues* 68, 1, 97–111. <https://doi.org/10.1111/j.1540-4560.2011.01738.x>
- [42] Andrew Crabtree, Tom Rodden, Peter Tolmie, and Graham Button. 2009. Ethnography considered harmful. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ’09)*, 879–888. <https://doi.org/10.1145/1518701.1518835>
- [43] Jenna Cryan, Shiliang Tang, Xinyi Zhang, Miriam Metzger, Haitao Zheng, and Ben Y. Zhao. 2020. Detecting gender stereotypes: Lexicon vs. Supervised learning methods. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI ’20)*, 1–11. <https://doi.org/10.1145/3313831.3376488>
- [44] Andreea Danielescu and Gwen Christian. 2018. A Bot is Not a Polyglot: Designing Personalities for Multi-Lingual Conversational Agents. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–9. <https://doi.org/10.1145/3170427.3174366>
- [45] Maitraye Das, Brent Hecht, and Darren Gergle. 2019. The Gendered Geography of Contributions to OpenStreetMap: Complexities in Self-Focus Bias. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3290605.3300793>
- [46] Shannon N. Davis and Theodore N. Greenstein. 2009. Gender Ideology: Components, Predictors, and Consequences. *Annual Review of Sociology* 35, 87–105.
- [47] Audrey Desjardins, Ron Wakkary, and William Odom. 2015. Investigating Genres and Perspectives in HCI Research on the Home. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 3073–3082. <https://doi.org/10.1145/2702123.2702540>
- [48] Patricia G. Devine. 1989. Stereotypes and prejudice: Their automatic and controlled components. *Journal of Personality and Social Psychology* 56, 1, 5–18. <https://doi.org/10.1037/0022-3514.56.1.5>
- [49] Catherine D’Ignazio and Lauren F. Klein. 2020. *Data Feminism*. MIT Press.
- [50] Tilman Dingler, Benjamin Tag, David A. Eccles, Niels van Berkel, and Vassilis Kostakos. 2022. Method for Appropriating the Brief Implicit Association Test to Elicit Biases in Users. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3517570>
- [51] Bryan Dosono and Bryan Semaan. 2018. Identity Work as Deliberation: AAPI Political Discourse in the 2016 US Presidential Election. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3173574.3174207>
- [52] Bryan Dosono, Bryan Semaan, and Jeff Hemsley. 2017. Exploring AAPI Identity Online: Political Ideology as a Factor Affecting Identity Work on Reddit. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 2528–2535. <https://doi.org/10.1145/3027063.3053185>
- [53] Paul Dourish. 2010. HCI and environmental sustainability: The politics of design and the design of politics. In *Proceedings of the 8th ACM Conference on Designing Interactive Systems (DIS ’10)*, 1–10. <https://doi.org/10.1145/1858171.1858173>
- [54] Benjamin J. Drury and Cheryl R. Kaiser. 2014. Allies against Sexism: The Role of Men in Confronting Sexism. *Journal of Social Issues* 70, 4, 637–652. <https://doi.org/10.1111/josi.12083>
- [55] Ron Eglash. 2002. Race, Sex, and Nerds: FROM BLACK GEEKS TO ASIAN AMERICAN HIPSTERS. *Social Text* 20, 2 (71), 49–64. https://doi.org/10.1215/01642472-20-2_71-49
- [56] Karla Elliott. 2016. Caring Masculinities: Theorizing an Emerging Concept. *Men and Masculinities* 19, 3, 240–259. <https://doi.org/10.1177/1097184X15576203>
- [57] Nathan Ensmenger. 2015. “Beards, Sandals, and Other Signs of Rugged Individualism”: Masculine Culture within the Computing Professions. *Osiris* 30, 1, 38–65. <https://doi.org/10.1086/682955>
- [58] Hayley Evans, Udaya Lakshmi, Hue Watson, Azra Ismail, Andrew M. Sherrill, Neha Kumar, and Rosa I. Arriaga. 2020. Understanding the Care Ecologies of Veterans with PTSD. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–15. <https://doi.org/10.1145/3313831.3376170>
- [59] Tracie Farrell, Miriam Fernandez, Jakub Novotny, and Harith Alani. 2019. Exploring misogyny across the manosphere in Reddit. In *Proceedings of the 10th ACM Conference on Web Science (WebSci ’19)*, 87–96. <https://doi.org/10.1145/3292522.3326045>
- [60] Anne Fausto-Sterling. 2000. *Sexing the Body: Gender Politics and the Construction of Sexuality*. Basic Books, New York, NY.
- [61] Andrew T. Fiore, Lindsay Shaw Taylor, G.A. Mendelsohn, and Marti Hearst. 2008. Assessing Attractiveness in Online Dating Profiles. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 797–806. <https://doi.org/10.1145/1357054.1357181>
- [62] Azadeh Forghani and Carman Neustaetter. 2014. The Routines and Needs of Grandparents and Parents for Grandparent-Grandchild Conversations over Distance. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 4177–4186. <https://doi.org/10.1145/2556288.2557255>
- [63] Dawn Foster. 2016. Lean Out: Dawn Foster Asks: If One Percent Are Leaning in, What Are the Other 99% Supposed to Do? Duncan Baird Publishers.
- [64] Jordan Foster and Jayne Baker. 2022. Muscles, makeup, and femboys: Analyzing TikTok’s “radical” masculinities. *Social Media & Society* 8, 3, 20563051221126040. <https://doi.org/10.1177/20563051221126040>
- [65] Jesse Fox and Wai Yen Tang. 2014. Sexism in online video games: The role of conformity to masculine norms and social dominance orientation. *Computers in Human Behavior* 33, 314–320. <https://doi.org/10.1016/j.chb.2013.07.014>
- [66] Sarah E. Fox, Meredith Lampe, and Daniela K. Rosner. 2018. Parody in Place: Exposing Socio-Spatial Exclusions in Data-Driven Maps with Design Parody. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3173574.3173896>
- [67] Sarah Fox and Daniela K. Rosner. 2016. Continuing the Dialogue: Bringing Research Accounts Back into the Field. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 1426–1430. <https://doi.org/10.1145/2858036.2858054>
- [68] Guo Freeman, Divine Maloney, Dane Acena, and Catherine Barwulor. 2022. (Re)Discovering the Physical Body Online: Strategies and Challenges to Approach Non-Cisgender Identity in Social Virtual Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3502082>
- [69] Sophie Freud. 1994. The social construction of gender. *Journal of Adult Development* 1, 1, 37–45. <https://doi.org/10.1007/BF02252981>
- [70] Daniel L. Gardner and Theresa Jean Tanenbaum. 2018. Dynamic Demographics: Lessons from a Large-Scale Census of Performative Possibilities in Games. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3173574.3173667>

- [71] Thomas J. Gerschick. 2011. *Disability Identity Intersections with Masculinities*. Routledge. <https://doi.org/10.4324/9780203833056-12>
- [72] Elisa Giaccardi and Elvin Karana. 2015. Foundations of Materials Experience: An Approach for HCI. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2447–2456. <https://doi.org/10.1145/2702123.2702337>
- [73] Debbie Ging. 2019. Alphas, betas, and incels: Theorizing the masculinities of the manosphere. *Men and Masculinities* 22, 4, 638–657. <https://doi.org/10.1177/1097184X17706401>
- [74] C. Gluck. 2022. What is Muscle Dysmorphia, Bigorexia, Reverse Anorexia? *HealthyPlace*. Retrieved January 16, 2023 from <https://www.healthyplace.com/ocd-related-disorders/body-dysmorphic-disorder/what-is-muscle-dysmorphia-bigorexia-reverse-anorexia>
- [75] Erving Goffman. 1959. The presentation of self in everyday life. In *Contemporary Sociological Theory*, Craig Calhoun, Joseph Gerteis, James Moody, Steven Pfaff and Indermohan Virk (eds.). John Wiley & Sons.
- [76] Amy Gonzales and Nicole Fritz. 2017. Prioritizing Flexibility and Intangibles: Medical Crowdfunding for Stigmatized Individuals. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 2371–2375. <https://doi.org/10.1145/3025453.3025647>
- [77] Jessica J. Good, Julie A. Woodzicka, Kimberly A. Bourne, and Corinne A. Moss-Racusin. 2019. 3 - The decision to act: Factors that predict women's and men's decisions to confront sexism. In *Confronting Prejudice and Discrimination*, Robyn K. Mallett and Margo J. Monteith (eds.). Academic Press, 49–71. <https://doi.org/10.1016/B978-0-12-814715-3.00003-5>
- [78] Shathel Haddad, Joanna McGrenere, and Claudia Jacova. 2014. Interface Design for Older Adults with Varying Cultural Attitudes toward Uncertainty. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1913–1922. <https://doi.org/10.1145/2556288.2557124>
- [79] Oliver L. Haimson, Anne E. Bowser, Edward F. Melcer, and Elizabeth F. Churchill. 2015. Online Inspiration and Exploration for Identity Reinvention. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 3809–3818. <https://doi.org/10.1145/2702123.2702270>
- [80] Oliver L. Haimson, Dyke Gorrell, Denny L. Starks, and Zu Weinger. 2020. Designing trans technology: Defining challenges and envisioning community-centered solutions. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI '20), 1–13. <https://doi.org/10.1145/3313831.3376669>
- [81] Foad Hamidi, Morgan Klaus Scheuerman, and Stacy M. Branham. 2018. Gender recognition or gender reductionism? The social implications of embedded gender recognition systems. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (CHI '18), 1–13. <https://doi.org/10.1145/3173574.3173582>
- [82] Sabine Harrer, Simon Nielsen, and Patrick Jarnfelt. 2019. Of Mice and Pants: Queering the Conventional Gamer Mouse for Cooperative Play. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–11. <https://doi.org/10.1145/3290607.3310431>
- [83] Christina N. Harrington, Shamika Klassen, and Yolanda A. Rankin. 2022. "All That You Touch, You Change": Expanding the Canon of Speculative Design Towards Black Futuring. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3502118>
- [84] Deneen M. Hatmaker. 2013. Engineering Identity: Gender and Professional Identity Negotiation among Women Engineers. *Gender, Work & Organization* 20, 4, 382–396. <https://doi.org/10.1111/j.1468-0432.2012.00589.x>
- [85] Alan J. Hawkins, Kay P. Bradford, Rob Palkovitz, Shawn L. Christiansen, Randal D. Day, and Vaughn R. A. Call. 2002. The Inventory of Father Involvement: A pilot study of a new measure of father involvement. *The Journal of Men's Studies* 10, 183–196. <https://doi.org/10.3149/jms.1002.183>
- [86] Mar Hicks. 2013. De-Programming the History of Computing [Think Piece]. *IEEE Annals of the History of Computing* 35, 1, 88–88. <https://doi.org/10.1109/MAHC.2013.3>
- [87] Julian Higgins, James Thomas, Jacqueline Chandler, Miranda Cumpston, Tian-jing Li, Matthew Page, and Vivian Welch. *Cochrane Handbook for Systematic Reviews of Interventions*. Retrieved October 1, 2021 from <https://training.cochrane.org/handbook/current>
- [88] Charles G. Hill, Maren Haag, Alannah Oleson, Chris Mendez, Nicola Marsden, Anita Sarma, and Margaret Burnett. 2017. Gender-Inclusiveness Personas vs. Stereotyping: Can We Have It Both Ways? In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 6658–6671. <https://doi.org/10.1145/3025453.3025609>
- [89] Julia Himmelsbach, Stephanie Schwarz, Cornelia Gerdenitsch, Beatrix Wais-Zechmann, Jan Bobeth, and Manfred Tscheli. 2019. Do We Care About Diversity in Human Computer Interaction: A Comprehensive Content Analysis on Diversity Dimensions in Research. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–16. <https://doi.org/10.1145/3290605.3300720>
- [90] Hofstede Insights. 2017. Country comparison. *Hofstede Insights*. Retrieved January 15, 2023 from <https://www.hofstede-insights.com/country-comparison/>
- [91] bell hooks. 2000. *Feminism Is for Everybody: Passionate Politics*. Pluto Press.
- [92] bell hooks. 2013. Dig Deep: Beyond Lean In. *The Feminist Wire*. Retrieved January 14, 2023 from <https://thefeministwire.com/2013/10/17973/>
- [93] Evan Hoovler. 2010. The male gamer stereotype dissected. *GameSpy*. Retrieved April 18, 2012 from <http://www.gamespy.com/articles/113/1136797p1.html>
- [94] Lara Houston, Steven J. Jackson, Daniela K. Rosner, Syed Ishtiaque Ahmed, Meg Young, and Laewoo Kang. 2016. Values in Repair. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 1403–1414. <https://doi.org/10.1145/2858036.2858470>
- [95] Lilian Hunt, Mathias Wullum Nielsen, and Londa Schiebinger. 2022. A framework for sex, gender, and diversity analysis in research. *Science* 377, 6614, 1492–1495. <https://doi.org/10.1126/science.abp9775>
- [96] Janet Shibley Hyde, Rebecca S. Bigler, Daphna Joel, Charlotte Chucky Tate, and Sari M. van Anders. 2019. The future of sex and gender in psychology: Five challenges to the gender binary. *American Psychologist* 74, 2, 171–193. <https://doi.org/10.1037/amp0000307>
- [97] Margaret Jack, Jay Chen, and Steven J. Jackson. 2017. Infrastructure as Creative Action: Online Buying, Selling, and Delivery in Phnom Penh. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 6511–6522. <https://doi.org/10.1145/3025453.3025889>
- [98] Betsy James DiSalvo, Sarita Yardi, Mark Gudzial, Tom McKlin, Charles Meadows, Kenneth Perry, and Amy Bruckman. 2011. African American Men Constructing Computing Identity. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2967–2970. <https://doi.org/10.1145/1978942.1979381>
- [99] Samantha Jaroszewski, Danielle Lottridge, Oliver L. Haimson, and Katie Quehl. 2018. "Genderfluid" or "attack helicopter": Responsible HCI research practice with non-binary gender variation in online communities. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (CHI '18), 1–15. <https://doi.org/10.1145/3173574.3173881>
- [100] Sheila Jeffreys. 2008. Keeping Women Down and Out: The Strip Club Boom and the Reinforcement of Male Dominance. *Signs: Journal of Women in Culture and Society* 34, 1, 151–173. <https://doi.org/10.1086/588501>
- [101] Robin Johnson. 2018. Technomascularity and Its Influence in Video Game Production. In *Masculinities in Play*, Nicholas Taylor and Gerald Voorhees (eds.). Springer International Publishing, Cham, 249–262. https://doi.org/10.1007/978-3-319-90581-5_14
- [102] Eun Hwa Jung, T. Franklin Waddell, and S. Shyam Sundar. 2016. Feminizing robots: User responses to gender cues on robot body and screen. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (CHI EA '16), 3107–3113. <https://doi.org/10.1145/2851581.2892428>
- [103] Dominic Kao, Rabindra Ratan, Christos Mousas, Amogh Joshi, and Edward F. Melcer. 2022. Audio Matters Too: How Audial Avatar Customization Enhances Visual Avatar Customization. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3501848>
- [104] Evangelos Karapanos, Stephan Wensveen, Bart Friederichs, and Jean-Bernard Martens. 2008. Do Knobs Have Character? Exploring Diversity in Users' Inferences. In *CHI '08 Extended Abstracts on Human Factors in Computing Systems*, 2907–2912. <https://doi.org/10.1145/1358628.1358782>
- [105] Lori Kendall. 2000. "Oh No! I'm a Nerd!": Hegemonic Masculinity on an Online Forum. *Gender and Society* 14, 2, 256–274.
- [106] Michael S. Kimmel. 1997. Masculinity as homophobia: Fear, shame and silence in the construction of gender identity. In *Toward a New Psychology of Gender*, M. M. Gergen and S. N. Davis (eds.). Taylor & Francis/Routledge, Florence, KY, US, 223–242.
- [107] Michael S. Kimmel, Jeff Hearn, and Robert W. Connell. 2004. *Handbook of Studies on Men and Masculinities*. SAGE Publications.
- [108] Agnieszka Kitkowska, Johan Högborg, and Erik Wästlund. 2022. Online Terms and Conditions: Improving User Engagement, Awareness, and Satisfaction through UI Design. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3517720>
- [109] Sandeep Kaur Kuttal, Bali Ong, Kate Kwasny, and Peter Robe. 2021. Trade-Offs for Substituting a Human with an Agent in a Pair Programming Context: The Good, the Bad, and the Ugly. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445659>
- [110] Xingyu Lan, Yanqiu Wu, Yang Shi, Qing Chen, and Nan Cao. 2022. Negative Emotions, Positive Outcomes? Exploring the Communication of Negativity in Serious Data Stories. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3517530>
- [111] Eun Ju Lee, Clifford Nass, and Scott Brave. 2000. Can computer-generated speech have gender?: An experimental test of gender stereotype. In *Proceedings of the CHI '00 Extended Abstracts on Human factors in Computing Systems* (CHI EA '00), 289–290. <https://doi.org/10.1145/633292.633461>
- [112] Minha Lee, Renee Noortman, Cristina Zaga, Alain Starke, Gijs Huisman, and Kristina Andersen. 2021. Conversational Futures: Emancipating Conversational Interactions for Futures Worth Wanting. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445244>
- [113] Sky Leslie, Mirjam Vosmeer, Casper Sterrenburg, Anastasia Maimenscu, Damir Catibovic, and Olico Matsjitadze. 2022. VR for Diversity: A Virtual Museum Exhibition about LGBTQ+. In *Extended Abstracts of the 2022 CHI Conference on*

- Human Factors in Computing Systems*. <https://doi.org/10.1145/3491101.3519917>
- [114] Weiwen Leung, Zheng Zhang, Daviti Jibuti, Jinhao Zhao, Maximilian Klein, Casey Pierce, Lionel Robert, and Haiyi Zhu. 2020. Race, Gender and Beauty: The Effect of Information Provision on Online Hiring Biases. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–11. <https://doi.org/10.1145/3313831.3376874>
- [115] Yi-Chi Liao, Kashyap Todi, Aditya Acharya, Antti Keurulainen, Andrew Howes, and Antti Oulasvirta. 2022. Rediscovering Affordance: A Reinforcement Learning Perspective. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3501992>
- [116] Merete Lie. 1995. Technology and masculinity: The case of the computer. *European Journal of Women's Studies* 2, 3, 379–394. <https://doi.org/10.1177/135050689500200306>
- [117] Chong-U Lim and D. Fox Harrell. 2015. Toward Telemetry-Driven Analytics for Understanding Players and Their Avatars in Videogames. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*, 1175–1180. <https://doi.org/10.1145/2702613.2732783>
- [118] Cindy Lin and Silvia Margot Lindtner. 2021. Techniques of Use: Confronting Value Systems of Productivity, Progress, and Usefulness in Computing and Design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445237>
- [119] Yu-Wei Lin and Matthijs den Besten. 2019. Gendered work culture in free/libre open source software development. *Gender, Work & Organization* 26, 7, 1017–1031. <https://doi.org/10.1111/gwao.12255>
- [120] Linda L. Lindsey. 2020. *Gender: Sociological Perspectives*. Routledge, New York. <https://doi.org/10.4324/9781315102023>
- [121] Sebastian Linxen, Christian Sturm, Florian Brühlmann, Vincent Cassau, Klaus Opwis, and Katharina Reinecke. 2021. How WEIRD is CHI? In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*, 1–14. <https://doi.org/10.1145/3411764.3445488>
- [122] John Lofland, David Snow, Leon Anderson, and Lyn H. Lofland. 2022. *Analyzing Social Settings: A Guide to Qualitative Observation and Analysis, Fourth Edition*. Waveland Press.
- [123] Sarah Lopez, Yi Yang, Kevin Beltran, Soo Jung Kim, Jennifer Cruz Hernandez, Chelsy Simran, Bingkun Yang, and Beste F. Yuksel. 2019. Investigating Implicit Gender Bias and Embodiment of White Males in Virtual Reality with Full Body Visuomotor Synchrony. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. <https://doi.org/10.1145/3290605.3300787>
- [124] Kai Lukoff, Carol Moser, and Sarita Schoenebeck. 2017. Gender Norms and Attitudes about Childcare Activities Presented on Father Blogs. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 4966–4971. <https://doi.org/10.1145/3025453.3025767>
- [125] Ning F. Ma, Veronica A. Rivera, Zheng Yao, and Dongwook Yoon. 2022. “Brush It Off”: How Women Workers Manage and Cope with Bias and Harassment in Gender-Agnostic Gig Platforms. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3517524>
- [126] Daniel Madden, Yuxuan Liu, Haowei Yu, Mustafa Feyyaz Sonbudak, Giovanni M Troiano, and Casper Harteveld. 2021. “Why Are You Playing Games? You Are a Girl!”: Exploring Gender Biases in Esports. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445248>
- [127] James R. Mahalik, Benjamin D. Locke, Larry H. Ludlow, Matthew A. Diemer, Ryan P. J. Scott, Michael Gottfried, and Gary Freitas. 2003. Development of the Conformity to Masculine Norms Inventory. *Psychology of Men & Masculinity* 4, 1, 3–25. <https://doi.org/10.1037/1524-9220.4.1.3>
- [128] Nicola Marsden. 2014. Doing Gender in Input Fields. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems*, 1399–1404. <https://doi.org/10.1145/2559206.2581212>
- [129] Nicola Marsden and Maren Haag. 2016. Stereotypes and Politics: Reflections on Personas. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 4017–4031. <https://doi.org/10.1145/2858036.2858151>
- [130] Nicola Marsden and Monika Probst. 2019. Personas and Identity: Looking at Multiple Identities to Inform the Construction of Personas. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3290605.3300565>
- [131] Jean-Bernard Martens. 2019. Interpreting the Diversity in Subjective Judgments. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–11. <https://doi.org/10.1145/3290605.3300449>
- [132] Mehreen Masood, Mujtaba Ahmed Khawaja, Muhammad Shehryaar Sharif, Omer Iqbal, Momin Mehmood Butt, and Suleman Shahid. 2021. Meri Kahani: A Gamified Solution to Teach Computational Thinking to Female Teenagers in Low Resource Communities. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411763.3451776>
- [133] Dan Maynes-Aminzade and Hayes Raffle. 2003. You're in Control: A Urinary User Interface. In *CHI '03 Extended Abstracts on Human Factors in Computing Systems*, 986–987. <https://doi.org/10.1145/765891.766108>
- [134] Dana McKay and Charlynn Miller. 2021. Standing in the Way of Control: A Call to Action to Prevent Abuse through Better Design of Smart Technologies. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445114>
- [135] Dana McKay, Huiwen Zhang, and George Buchanan. 2022. Who am I, and who are you, and who are we? A Scientometric Analysis of Gender and Geography in HCI. In *CHI Conference on Human Factors in Computing Systems*, 1–19. <https://doi.org/10.1145/3491102.3502106>
- [136] Amanda Menking, Ingrid Erickson, and Wanda Pratt. 2019. People Who Can Take It: How Women Wikipedians Negotiate and Navigate Safety. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3290605.3300702>
- [137] Danaë Metaxa-Kakavouli, Kelly Wang, James A. Landay, and Jeff Hancock. 2018. Gender-Inclusive Design: Sense of Belonging and Bias in Web Interfaces. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–6. <https://doi.org/10.1145/3173574.3174188>
- [138] Lydia Michie, Madeline Balaam, John McCarthy, Timur Osadchiy, and Kellie Morrissey. 2018. From Her Story, to Our Story: Digital Storytelling as Public Engagement around Abortion Rights Advocacy in Ireland. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–15. <https://doi.org/10.1145/3173574.3173931>
- [139] Alfredo Mirandé. 2016. Hombres mujeres: An Indigenous third gender. *Men and Masculinities* 19, 4, 384–409. <https://doi.org/10.1177/1097184X15602746>
- [140] Preeti Mudliar and Nimmi Rangaswamy. 2015. Offline Strangers, Online Friends: Bridging Classroom Gender Segregation with WhatsApp. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 3799–3808. <https://doi.org/10.1145/2702123.2702533>
- [141] Zachary Munn, Micah D. J. Peters, Cindy Stern, Catalin Tufanaru, Alexa McArthur, and Edoardo Aromataris. 2018. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Medical Research Methodology* 18, 1, 143. <https://doi.org/10.1186/s12874-018-0611-x>
- [142] Tyler Musgrave, Alia Cummings, and Sarita Schoenebeck. 2022. Experiences of Harm, Healing, and Joy among Black Women and Femmes on Social Media. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3517608>
- [143] Elizabeth D. Mynatt, Jim Rowan, Sarah Craighill, and Annie Jacobs. 2001. Digital Family Portraits: Supporting Peace of Mind for Extended Family Members. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 333–340. <https://doi.org/10.1145/365024.365126>
- [144] Jennifer C. Ng, Sharon S. Lee, and Yoon K. Pak. 2007. Chapter 4 Contesting the Model Minority and Perpetual Foreigner Stereotypes: A Critical Review of Literature on Asian Americans in Education. *Review of Research in Education* 31, 1, 95–130. <https://doi.org/10.3102/0091732X07300046095>
- [145] Anna Offenwanger, Alan John Milligan, Minsuk Chang, Julia Bullard, and Dongwook Yoon. 2021. Diagnosing bias in the gender representation of HCI research participants: How it happens and where we are. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*, 1–18. <https://doi.org/10.1145/3411764.3445383>
- [146] Johanna Okerlund, David Wilson, and Celine Latulipe. 2021. A Feminist Utopian Perspective on the Practice and Promise of Making. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445126>
- [147] Lisa Orii, Nami Ogawa, Yuji Hatada, and Takuji Narumi. 2022. Designing for Speech Practice Systems: How Do User-Controlled Voice Manipulation and Model Speakers Impact Self-Perceptions of Voice? In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3502093>
- [148] Jahnna Otterbacher. 2015. Crowdsourcing Stereotypes: Linguistic Bias in Metadata Generated via GWAP. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 1955–1964. <https://doi.org/10.1145/2702123.2702151>
- [149] Jahnna Otterbacher, Jo Bates, and Paul Clough. 2017. Competent Men and Warm Women: Gender Stereotypes and Backlash in Image Search Results. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 6620–6631. <https://doi.org/10.1145/3025453.3025727>
- [150] Tyler Pace. 2008. Can an Orc Catch a Cab in Stormwind? Cybertype Preference in the World of Warcraft Character Creation Interface. In *CHI '08 Extended Abstracts on Human Factors in Computing Systems*, 2493–2502. <https://doi.org/10.1145/1358628.1358706>
- [151] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, Roger Chou, Julie Glanville, Jeremy M Grimshaw, Asbjørn Hróbjartsson, Manoj M Lalu, Tianjing Li, Elizabeth W Loder, Evan Mayo-Wilson, Steve McDonald, Luke A McGuinness, Lesley A Stewart, James Thomas, Andrea C Tricco, Vivian A Welch, Penny Whiting, and David Moher. 2021. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, n71. <https://doi.org/10.1136/bmj.n71>
- [152] C. J. Pascoe and Tristan Bridges. 2016. *Exploring Masculinities: Identity, Inequality, Continuity and Change*. Oxford University Press.

- [153] Jessica A. Pater, Lauren E. Reining, Andrew D. Miller, Tammy Toscos, and Elizabeth D. Mynatt. 2019. "Notjustgirls": Exploring Male-Related Eating Disordered Content across Social Media Platforms. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1-13. <https://doi.org/10.1145/3290605.3300881>
- [154] Sachin R Pendse, Daniel Nkemelu, Nicola J Bidwell, Sushrut Jadhav, Soumitra Pathare, Munmun De Choudhury, and Neha Kumar. 2022. From Treatment to Healing: Envisioning a Decolonial Digital Mental Health. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3501982>
- [155] Giulia Perugia, Stefano Guidi, Margherita Bicchi, and Oronzo Parlange. 2022. The shape of our bias: Perceived age and gender in the humanoid robots of the ABOT database. In *Proceedings of the 2022 ACM/IEEE International Conference on Human-Robot Interaction (HRI '22)*, 110-119.
- [156] Micah D. J. Peters, Christina M. Godfrey, Hanan Khalil, Patricia McInerney, Deborah Parker, and Cassia Baldini Soares. 2015. Guidance for conducting systematic scoping reviews. *International Journal of Evidence-Based Healthcare* 13, 3, 141-146. <https://doi.org/10.1097/XEB.0000000000000050>
- [157] Katharine A. Phillips. 2009. *Understanding Body Dysmorphic Disorder*. Oxford University Press.
- [158] Roosa Piitulainen, Perttu Hämäläinen, and Elisa D Mekler. 2022. Vibing Together: Dance Experiences in Social Virtual Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3501828>
- [159] Liz Plank. 2019. *For the Love of Men: From Toxic to a More Mindful Masculinity*. St. Martin's Press.
- [160] Joseph H. Pleck. 1981. *The Myth of Masculinity*. MIT Press.
- [161] Joseph H. Pleck. 1995. The gender role strain paradigm: An update. In *A new psychology of men*. Basic Books/Hachette Book Group, New York, NY, US, 11-32.
- [162] Jie Qi, Leah Buechley, Andrew "bunnie" Huang, Patricia Ng, Sean Cross, and Joseph A. Paradiso. 2018. Chibitronics in the Wild: Engaging New Communities in Creating Technology with Paper Electronics. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1-11. <https://doi.org/10.1145/3173574.3173826>
- [163] Dennis E. Reidy, Joanne P. Smith-Darden, Kai S. Cortina, Roger M. Kernsmith, and Poco D. Kernsmith. 2015. Masculine Discrepancy Stress, Teen Dating Violence, and Sexual Violence Perpetration Among Adolescent Boys. *Journal of Adolescent Health* 56, 6, 619-624. <https://doi.org/10.1016/j.jadohealth.2015.02.009>
- [164] Jennifer A. Rode. 2011. Reflexivity in digital anthropology. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*, 123-132. <https://doi.org/10.1145/1978942.1978961>
- [165] Jennifer A. Rode. 2011. A theoretical agenda for feminist HCI. *Interacting with Computers* 23, 5, 393-400. <https://doi.org/10.1016/j.intcom.2011.04.005>
- [166] Daniela K. Rosner. 2014. Making Citizens, Reassembling Devices: On Gender and the Development of Contemporary Public Sites of Repair in Northern California. *Public Culture* 26, 1 (72), 51-77. <https://doi.org/10.1215/08992363-2346250>
- [167] Daniela K. Rosner, Samantha Shorey, Brock R. Craft, and Helen Remick. 2018. Making Core Memory: Design Inquiry into Gendered Legacies of Engineering and Craftwork. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1-13. <https://doi.org/10.1145/3173574.3174105>
- [168] Jennifer D. Rubin, Lindsay Blackwell, and Terri D. Conley. 2020. Fragile masculinity: Men, gender, and online harassment. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*, 1-14. <https://doi.org/10.1145/3313831.3376645>
- [169] Bonnie Ruder, Dwaine Plaza, Rebecca Warner, and Michelle Bothwell. 2018. STEM Women Faculty Struggling for Recognition and Advancement in a "Men's Club" Culture. In *Exploring the Toxicity of Lateral Violence and Microaggressions: Poison in the Water Cooler*, Christine L. Cho, Julie K. Corkett and Astrid Steele (eds.). Springer International Publishing, Cham, 121-149. https://doi.org/10.1007/978-3-319-74760-6_7
- [170] Cosima Rughiniș, Razvan Rughiniș, and Bogdana Humă. 2016. Impromptu Crowd Science and the Mystery of the Bechdel-Wallace Test Movement. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 487-500. <https://doi.org/10.1145/2851581.2892580>
- [171] Patricia Russo and Stephen Boor. 1993. How Fluent is Your Interface? Designing for International Users. In *Proceedings of the INTERACT '93 and CHI '93 Conference on Human Factors in Computing Systems*, 342-347. <https://doi.org/10.1145/169059.169274>
- [172] Anastasia Salter and Bridget Blodgett. 2012. Hypermasculinity & Dickwolves: The Contentious Role of Women in the New Gaming Public. *Journal of Broadcasting & Electronic Media* 56, 3, 401-416. <https://doi.org/10.1080/08838151.2012.705199>
- [173] Nithya Sambasivan, Amna Batool, Nova Ahmed, Tara Matthews, Kurt Thomas, Laura Sanely Gaytán-Lugo, David Nemer, Elie Bursztin, Elizabeth Churchill, and Sunny Consolvo. 2019. "They Don't Leave Us Alone Anywhere We Go": Gender and Digital Abuse in South Asia. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1-14. <https://doi.org/10.1145/3290605.3300232>
- [174] Sheryl Sandberg. 2015. Lean in - Women, Work and the Will to Lead. *NHRD Network Journal* 8, 2, 137-139. <https://doi.org/10.1177/0974173920150225>
- [175] Yukiko Sawaya, Mahmood Sharif, Nicolas Christin, Ayumu Kubota, Akihiro Nakarai, and Akira Yamada. 2017. Self-Confidence Trumps Knowledge: A Cross-Cultural Study of Security Behavior. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 2202-2214. <https://doi.org/10.1145/3025453.3025926>
- [176] Stuart Schechter, Serge Egelman, and Robert W. Reeder. 2009. It's Not What You Know, but Who You Know: A Social Approach to Last-Resort Authentication. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1983-1992. <https://doi.org/10.1145/1518701.1519003>
- [177] Morgan Klaus Scheuerman, Aaron Jiang, Katta Spiel, and Jed R. Brubaker. 2021. Revisiting Gendered Web Forms: An Evaluation of Gender Inputs with (Non-)Binary People. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445742>
- [178] Ari Schlesinger, W. Keith Edwards, and Rebecca E. Grinter. 2017. Intersectional HCI: Engaging identity through gender, race, and class. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*, 5412-5427. <https://doi.org/10.1145/3025453.3025766>
- [179] Stephen A. Schullo and Burton L. Alpers. 1984. Interpersonal phenomenology as a function of sexual orientation, sex, sentiment, and trait categories in long-term dyadic relationships. *Journal of Personality and Social Psychology* 47, 5, 983-1002. <https://doi.org/10.1037/0022-3514.47.5.983>
- [180] Katie Seaborn and Alexa Frank. 2022. What pronouns for Pepper? A critical review of gender/ing in research. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*, 1-15. <https://doi.org/10.1145/3491102.3501996>
- [181] Katie Seaborn, Peter Pennefather, and Haruki Kotani. 2022. Exploring gender-expansive categorization options for robots. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22)*, 1-6. <https://doi.org/10.1145/3491101.3519646>
- [182] Katie Seaborn and Jacqueline Urakami. 2021. Measuring voice UX quantitatively: A rapid review. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (CHI EA '21)*, No. 416. <https://doi.org/10.1145/3411763.3451712>
- [183] Orit Shaer, Lauren Westendorf, Nicholas A. Knouf, and Claudia Pederson. 2017. Understanding Gaming Perceptions and Experiences in a Women's College Community. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 1544-1557. <https://doi.org/10.1145/3025453.3025623>
- [184] Andra Siibak. 2010. Constructing masculinity on a social networking site: The case-study of visual self-presentations of young men on the profile images of SNS Rate. *YOUNG* 18, 4, 403-425. <https://doi.org/10.1177/110330881001800403>
- [185] Marie Louise Juul Søndergaard, Gopinath Kannabiran, Simran Chopra, Nadia Campo Woytuk, Dilrukshi Gamage, Ebtisam Alabdulqader, Heather McKinnon, Heike Winschiers-Theophilus, and Shaowen Bardzell. 2022. Feminist voices about ecological issues in HCI. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22)*, 1-7. <https://doi.org/10.1145/3491101.3503717>
- [186] Janet T. Spence, Robert Helmreich, and Joy Stapp. 1975. Ratings of self and peers on sex role attributes and their relation to self-esteem and conceptions of masculinity and femininity. *Journal of Personality and Social Psychology* 32, 29-39. <https://doi.org/10.1037/h0076857>
- [187] Katta Spiel, Oliver L. Haimson, and Danielle Lottridge. 2019. How to do better with gender on surveys: A guide for HCI researchers. *Interactions* 26, 4, 62-65. <https://doi.org/10.1145/3338283>
- [188] Katta Spiel, Os Keyes, and Pinar Barlas. 2019. Patching gender: Non-binary utopias in HCI. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems (CHI EA '19)*, 1-11. <https://doi.org/10.1145/3290607.3310425>
- [189] Derek Stanovsky. 2007. Postcolonial masculinities. *International encyclopedia of men and masculinities*, 493-496.
- [190] Yolande Strengers, Jenny Kennedy, Paula Arcari, Larissa Nicholls, and Melissa Gregg. 2019. Protection, Productivity and Pleasure in the Smart Home: Emerging Expectations and Gendered Insights from Australian Early Adopters. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1-13. <https://doi.org/10.1145/3290605.3300875>
- [191] Simone Stumpf, Anicia Peters, Shaowen Bardzell, Margaret Burnett, Daniela Busse, Jessica Cauchard, and Elizabeth Churchill. 2020. Gender-inclusive HCI research and design: A conceptual review. *Foundations and Trends® in Human-Computer Interaction* 13, 1, 1-69. <https://doi.org/10.1561/11000000056>
- [192] Sharifa Sultana, François Guimbretière, Phoebe Sengers, and Nicola Dell. 2018. Design Within a Patriarchal Society: Opportunities and Challenges in Designing for Rural Women in Bangladesh. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1-13. <https://doi.org/10.1145/3173574.3174110>
- [193] Jiao Sun, Tongshuang Wu, Yue Jiang, Ronil Awalegaonkar, Xi Victoria Lin, and Diyi Yang. 2022. Pretty Princess vs. Successful Leader: Gender Roles in Greeting Card Messages. In *Proceedings of the 2022 CHI Conference on Human Factors in*

- Computing Systems*. <https://doi.org/10.1145/3491102.3502114>
- [194] JaYoung Sung, Rebecca E. Grinter, and Henrik I. Christensen. 2009. "Pimp My Roomba": Designing for Personalization. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 193–196. <https://doi.org/10.1145/1518701.1518732>
- [195] Selina Jeanne Sutton, Paul Foulkes, David Kirk, and Shaun Lawson. 2019. Voice as a Design Material: Sociophonetic Inspired Design Strategies in Human-Computer Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3290605.3300833>
- [196] Franziska Tachtler, Reem Talhouk, Toni Michel, Petr Slovak, and Geraldine Fitzpatrick. 2021. Unaccompanied Migrant Youth and Mental Health Technologies: A Social-Ecological Approach to Understanding and Designing. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445470>
- [197] Ana Tajadura-Jiménez, Joseph Newbold, Linge Zhang, Patricia Rick, and Nadia Bianchi-Berthouze. 2019. As Light as You Aspire to Be: Changing Body Perception with Sound to Support Physical Activity. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3290605.3300888>
- [198] Cara Tannenbaum, Robert P. Ellis, Friederike Eysel, James Zou, and Londa Schiebinger. 2019. Sex and gender analysis improves science and engineering. *Nature* 575, 7781, 137–146. <https://doi.org/10.1038/s41586-019-1657-6>
- [199] Nick Taylor, Ursula Hurley, and Philip Connolly. 2016. Making Community: The Wider Role of Makerspaces in Public Life. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 1415–1425. <https://doi.org/10.1145/2858036.2858073>
- [200] Divy Thakkar, Nithya Sambasivan, Purva Kulkarni, Pratap Kalenahalli Sudarshan, and Kentaro Toyama. 2018. The Unexpected Entry and Exodus of Women in Computing and HCI in India. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*, 1–12. <https://doi.org/10.1145/3173574.3173926>
- [201] Helen Thornham. 2008. "It's a boy thing": Gaming, gender, and geeks. *Feminist Media Studies* 8, 2, 127–142. <https://doi.org/10.1080/14680770801980505>
- [202] Alexandra To, Hillary Carey, Riya Shrivastava, Jessica Hammer, and Geoff Kaufman. 2022. Interactive Fiction Prototypes for Coping with Interpersonal Racism. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3502044>
- [203] Kashyap Todi, Gilles Bailly, Luis Leiva, and Antti Oulasvirta. 2021. Adapting User Interfaces with Model-Based Reinforcement Learning. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445497>
- [204] Andrea C. Tricco, Erin Lillie, Wasifa Zarin, Kelly K. O'Brien, Heather Colquhoun, Danielle Levac, David Moher, Micah D.J. Peters, Tanya Horsley, Laura Weeks, Susanne Hempel, Elie A. Akl, Christine Chang, Jessie McGowan, Lesley Stewart, Lisa Hartling, Adrian Aldcroft, Michael G. Wilson, Chantelle Garritty, Simon Lewin, Christina M. Godfrey, Marilyn T. Macdonald, Etienne V. Langlois, Karla Soares-Weiser, Jo Moriarty, Tammy Clifford, Özge Tunçalp, and Sharon E. Straus. 2018. PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Annals of Internal Medicine* 169, 7, 467–473. <https://doi.org/10.7326/M18-0850>
- [205] Giovanni Maria Troiano, Matthew Wood, and Casper Hartevelde. 2020. "And This, Kids, Is How I Met Your Mother": Consumerist, Mundane, and Uncanny Futures with Sex Robots. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–17. <https://doi.org/10.1145/3313831.3376598>
- [206] Elizabeth (Dori) Tunstall. 2013. Decolonizing design innovation: Design anthropology, critical anthropology, and indigenous knowledge. In *Design Anthropology: Theory and Practice*, Wendy Gunn, Ton Otto and Rachel Charlotte Smith (eds.). Bloomsbury, London, UK, 232–250.
- [207] Reinis Udris. 2014. Cyberbullying among high school students in Japan: Development and validation of the Online Disinhibition Scale. *Computers in Human Behavior* 41, 253–261. <https://doi.org/10.1016/j.chb.2014.09.036>
- [208] Aditya Vashistha, Abhinav Garg, Richard Anderson, and Agha Ali Raza. 2019. Threats, Abuses, Flirting, and Blackmail: Gender Inequity in Social Media Voice Forums. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3290605.3300302>
- [209] Theresa K. Vescio and Natasza Kosakowska-Berezecka. 2020. The Not So Subtle and Status Quo Maintaining Nature of Everyday Sexism. In *The Cambridge Handbook of the International Psychology of Women*, Diane F. Halpern and Fanny M. Cheung (eds.). Cambridge University Press, Cambridge, 205–220. Retrieved January 14, 2023 from <https://www.cambridge.org/core/books/cambridge-handbook-of-the-international-psychology-of-women/not-so-subtle-and-status-quo-maintaining-nature-of-everyday-sexism/9C9107C8E76A324C3CE27F2EF84F5BF9>
- [210] Lisa Wade and Myra Marx Ferree. 2019. *Gender: Ideas, Interactions, Institutions*. W. W. Norton, Incorporated.
- [211] Annika Waern, Elena Balan, and Kim Nevelsteen. 2012. Athletes and Street Acrobats: Designing for Play as a Community Value in Parkour. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 869–878. <https://doi.org/10.1145/2207676.2208528>
- [212] Judy Wajcman. 1991. Technology as masculine culture. In *Feminism confronts technology*. Penn State University Press, Pennsylvania.
- [213] Judy Wajcman. 1991. Patriarchy, Technology, and Conceptions of Skill. *Work and Occupations* 18, 1, 29–45. <https://doi.org/10.1177/0730888491018001002>
- [214] Judy Wajcman. 2010. Feminist theories of technology. *Cambridge Journal of Economics* 34, 1, 143–152. <https://doi.org/10.1093/cje/ben057>
- [215] Sylvia Walby. 1989. Theorising patriarchy. *Sociology* 23, 2, 213–234. <https://doi.org/10.1177/0038038589023002004>
- [216] Andrea Waling. 2019. Rethinking Masculinity Studies: Feminism, Masculinity, and Poststructural Accounts of Agency and Emotional Reflexivity. *The Journal of Men's Studies* 27, 1, 89–107. <https://doi.org/10.1177/1060826518782980>
- [217] MinJuan Wang, Sus Lundgren Lyckvi, and Fang Chen. 2016. Why and How Traffic Safety Cultures Matter When Designing Advisory Traffic Information Systems. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 2808–2818. <https://doi.org/10.1145/2858036.2858467>
- [218] Yi-Chia Wang, Moira Burke, and Robert E. Kraut. 2013. Gender, Topic, and Audience Response: An Analysis of User-Generated Content on Facebook. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 31–34. <https://doi.org/10.1145/2470654.2470659>
- [219] Zixuan Wang, Jiawen Huang, and Costa Fiammetta. 2021. Analysis of Gender Stereotypes for the Design of Service Robots: Case Study on the Chinese Catering Market. In *Designing Interactive Systems Conference 2021*, 1336–1344. <https://doi.org/10.1145/3461778.3462087>
- [220] Michael Rhys Morgan Ward. 2015. The Chameleonisation of Masculinity: Jimmy's Multiple Performances of a Working-Class Self. *Masculinities & Social Change* 4, 3, 215–240. <https://doi.org/10.17583/mcs.2015.1657>
- [221] Astrid Weiss, Anna Pillinger, Katta Spiel, and Sabine Zauchner-Studnicka. 2020. Inconsequential Appearances: An Analysis of Anthropomorphic Language in Voice Assistant Forums. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–7. <https://doi.org/10.1145/3334480.3382793>
- [222] Candace West and Don H. Zimmerman. 1987. Doing gender. *Gender & Society* 1, 2, 125–151. <https://doi.org/10.1177/0891243287001002002>
- [223] Rua M. Williams and Juan E. Gilbert. 2019. Cyborg perspectives on computing research reform. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–11. <https://doi.org/10.1145/3290607.3310421>
- [224] Langdon Winner. 2007. Do artifacts have politics? In *Computer Ethics*. Routledge. Retrieved from <https://doi.org/10.4324/9781315259697>
- [225] Emma Witkowski. 2012. Inside the huddle: The phenomenology and sociology of team play in networked computer games. *IT University of Copenhagen, Copenhagen*.
- [226] Emma Witkowski. 2013. *Eventful Masculinities: Negotiations of Hegemonic Sporting Masculinities at LANs*. Routledge. <https://doi.org/10.4324/9780203084496-20>
- [227] Matthew Wood, Gavin Wood, and Madeline Balaam. 2017. "They're Just Tixel Pits, Man": Disputing the 'Reality' of Virtual Reality Pornography through the Story Completion Method. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 5439–5451. <https://doi.org/10.1145/3025453.3025762>
- [228] Xuexin Xu, Yin-Leng Theng, Jinhui Li, and Pham Tan Phat. 2016. Investigating Effects of Exergames on Exercise Intentions among Young-Old and Old-Old. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 2961–2968. <https://doi.org/10.1145/2851581.2892296>
- [229] Judith Yaaqoubi and Katharina Reinecke. 2018. The Use and Usefulness of Cultural Dimensions in Product Development. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–9. <https://doi.org/10.1145/3170427.3174368>
- [230] Amal Yassien, ElHassan B. Makled, Passant Elagroudy, Nouran Sadek, and Slim Abdennadher. 2021. Give-Me-A-Hand: The Effect of Partner's Gender on Collaboration Quality in Virtual Reality. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411763.3451601>
- [231] Renwen Zhang, Natalya N. Bazarova, and Madhu Reddy. 2021. Distress Disclosure across Social Media Platforms during the COVID-19 Pandemic: Untangling the Effects of Platforms, Affordances, and Audiences. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3411764.3445134>
- [232] Andrew Zolides. Lipstick bullets: Labour and gender in professional gamer self-branding. *Persona Studies* 1, 2, 42–53. <https://doi.org/10.3316/inform.968534603571037>
- [233] 2018. Man. *The Oxford Dictionary*.
- [234] Katie Seaborn, Giulia Barbareschi, and Shruti Chandra. 2023. Not Only WEIRD but "Uncanny"? A Systematic Review of Diversity in Human-Robot Interaction Research. *International Journal of Social Robotics*, 1–30. <https://doi.org/10.1007/s12369-023-00968-4>
- [235] Jenny L. Davis. 2019. Refusing (mis)recognition: Navigating multiple marginalization in the U.S. Two Spirit movement. *Review of International American Studies* 12, 1, 65–86. <https://doi.org/10.31261/rias.7328>